- of Section S3.1 (`analysis/mpce_inference_extra.py`) and
- of Section S5.3 (`analysis/mpce_direction.py`);

the diagnostics of Section S7 come from instrumented reruns (`analysis/mpce_diagnostics.py`), and every number of Section S6 is recomputed by `analysis/make_theory_figures.py`.

S1 Evaluation Protocol and Implementation Details

This section gives the data, the pseudocode and the settings needed to reproduce the protocol of Section 7:

- the two wind data sets of the 68 benchmark cases, the salp-swarm methods compared as Phase-1 alternatives in the component analysis, the details of the ranking and the tests (Sections S1.1–S1.4), and
- the implementation details that the main text does not state: iteration counts, MS-SLSQP, the feasibility-preserving initialization, the Horns Rev 1 model and the feasibility tolerance (Sections S1.5–S1.9).

All settings are those of the code in `analysis/` (Section S9).

S1.1 Wind Data

Table S1 lists the two wind data sets of the Kusiak–Song benchmark [1, 2].

Table S1: Wind Data Sets I and II: probability p_l of direction bin l ($[15(l-1), 15l)^\circ$, direction the wind blows towards, counter-clockwise from east) and Weibull scale ψ; $\zeta = 2$ in all bins. The most probable bin is $l = 7$ (90°–105° towards, i.e., wind from 165°–180° in the meteorological convention $\theta_{\text{met}} = 270^\circ - \theta$) for Data Set I and $l = 13$ (180°–195° towards, i.e., wind from 75°–90°) for Data Set II.

Bin l	1	2	3	4	5	6	7	8	9	10	11	12
DS I: p_l ($\psi = 13$)	0.000	0.010	0.010	0.010	0.010	0.200	0.600	0.010	0.010	0.010	0.010	0.010
DS II: ψ	7.0	5.0	5.0	5.0	5.0	4.0	5.0	6.0	7.0	7.0	8.0	9.5
DS II: p_l	0.0002	0.0080	0.0227	0.0242	0.0225	0.0339	0.0423	0.0290	0.0617	0.0813	0.0994	0.1394
Bin l	13	14	15	16	17	18	19	20	21	22	23	24
DS I: p_l ($\psi = 13$)	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.000
DS II: ψ	10.0	8.5	8.5	6.5	4.6	2.6	8.0	5.0	6.4	5.2	4.5	3.9
DS II: p_l	0.1839	0.1115	0.0765	0.0080	0.0051	0.0019	0.0012	0.0010	0.0017	0.0031	0.0097	0.0317

S1.2 Model Details

This subsection gives the details of the model of Section 3 that the main text only summarizes.

Discretized expected power. The integral (3) is evaluated, as in [1], by a Riemann sum over direction bins $\theta_0 = 0^\circ, \theta_1, \ldots, \theta_{N_\theta} = 360^\circ$ with probabilities p_l, and over the speed range between $s_{\text{cut-in}}$ and s_{rated}, divided into N_s bins $s_0 = s_{\text{cut-in}}, s_1, \ldots, s_{N_s} = s_{\text{rated}}$ (benchmark: $N_\theta = 24$ bins of 15° and $N_s = 21$ bins of 0.5 m/s). With the bin centers $\bar{\theta}_l = (\theta_{l-1} + \theta_l)/2$ and the Weibull survival function $W_i(s, \theta) = \exp\{-[s/\psi_i(\theta)]^{\zeta(\theta)}\}$ of the local speed,

$$E(P_i) = \sum_{l=1}^{N_\theta} (\theta_l - \theta_{l-1})\, p_l \Bigg\{ P_{\text{rated}}\, W_i(s_{\text{rated}}, \bar{\theta}_l) + \lambda_1 \sum_{q=1}^{N_s} \tfrac{s_{q-1}+s_q}{2} \big[W_i(s_{q-1}, \bar{\theta}_l) - W_i(s_q, \bar{\theta}_l) \big] + \lambda_0 \big[W_i(s_{\text{cut-in}}, \bar{\theta}_l) - W_i(s_{\text{rated}}, \bar{\theta}_l) \big] \Bigg\}. \tag{S1}$$

The bin-width factor $(\theta_l - \theta_{l-1})$ is kept in degrees, as in the benchmark studies [1]; with 24 bins of 15° and $\sum_l p_l = 1$, the benchmark objective $\sum_i E(P_i)$ is 15 times the expected farm power in kW.

Power curve. We use it as published: it is slightly negative just above the cut-in speed (−7.0 kW at 3.5 m/s, zero at 3.55 m/s), jumps from 1472 to 1500 kW at the rated speed and has no cut-out speed. We keep this linearized curve for comparability with the published Kusiak–Song results; its effect on the energy values is quantified with a cubic curve in Section 8.5.1, and the Horns Rev 1 block uses the tabulated V80 power curve (Section 9).

Why the direction discretization matters. The objective evaluates the wake only at the 24 bin centers $\bar{\theta}_l = (15l - 7.5)^\circ$. A top-hat wake is narrow: by the half-cone geometry (Fig. 1c), turbine j wakes a turbine i at the distance $\ell_{ij} \geq R/K$ only for directions within an interval of width $2[\alpha + \arcsin(R/(\ell_{ij}\sqrt{1+K^2}))]$ around

the direction from j to i. With $K = 0.075$ and $R = 38.5$ m ($\alpha = 4.29°$, $R/K = 513$ m), this width is below the bin spacing of 15° for $\ell_{ij} > 685$ m and tends to $2\alpha = 8.6°$. If no bin center lies in this interval, the wake of the pair does not enter the objective (with bins of 5° or finer, every such interval contains a center). Optimized layouts exploit these gaps (with 1° sub-bins, the mean wake loss of PSO-VNS rises by about 1 pp); Section 8.5.5 reports which conclusions survive a 1° rose, for re-evaluated and for directly optimized layouts.

Why the Jensen model. First, the Jensen model with the benchmark settings is needed for comparability with the published Kusiak–Song results [1, 2], and it is cheap enough for the 49,010 optimization runs of this study (Table 4). Second, the research question concerns the optimizers and the protocol rather than the fidelity of the wake model; whether the conclusions carry over to another wake model is an empirical question. Third, we therefore test a Gaussian wake in two ways: the methods are re-optimized under the simplified Bastankhah Gaussian wake of IEA37 Case Study 1 (Section 11), and the final benchmark layouts are re-evaluated with the Gaussian wake of [13] (Section 8.5.2). Control-oriented models, which describe the wake deflection of yawed turbines for wake steering [15, 16], add yaw angles as decision variables and turn the problem into a layout–control co-design, which is outside the scope of this study.

S1.3 Salp Swarm Algorithm and Its Laplacian Variant

Algorithm S1 gives the pseudocode of SSA [3] and LX-SSA [4] as described in Section 4.2 of the main text, as used in Phase 1 of SSA-VNS and LX-SSA-VNS and as stand-alone methods; the evaluations per iteration are derived there, and the resulting iteration counts are given in Section S1.5.

Algorithm S1 SSA and LX-SSA (lines marked LX: LX-SSA only); evaluations per line in brackets

1: Draw and evaluate N_p salps; sort them by F_p and set the best as food source $\mathbf{H}$. ▷ $[N_p]$
2: **for** $t = 1$ to T **do**
3: Update $\xi_1 = 2\exp[-(4t/T)^2]$.
4: **for** salp $i = 1, \ldots, N_p$ **do**
5: **if** $i \le N_p/2$ **then**
6: Update leader i using (9) (new position $\tilde{\mathbf{X}}^i$).
7: **else**
8: Form the follower candidate $\mathbf{Z}^i = \frac{1}{2}(\mathbf{X}^i + \tilde{\mathbf{X}}^{i-1})$ and set $\tilde{\mathbf{X}}^i \leftarrow \mathbf{Z}^i$.
9: LX: form $\mathbf{Y}^i$ using (10), box-repair $\mathbf{Y}^i$ and $\mathbf{Z}^i$, evaluate both and set $\tilde{\mathbf{X}}^i$ to the better one. ▷ [2 per follower]
10: **end if**
11: **end for**
12: Box-repair $\tilde{\mathbf{X}}$, set $\mathbf{X} \leftarrow \tilde{\mathbf{X}}$, evaluate F_p for all salps, sort them, and update $\mathbf{H}$. ▷ $[N_p]$
13: **end for**
14: **return** $\mathbf{H}$.

Implementation details. SSA and LX-SSA are run as in the code of the published LX-SSA study [4], which departs from the description of Mirjalili et al. [3] in three points:

(a) the paper defines a single leader, the first salp, and followers $i \ge 2$, whereas the implementation treats the first $N_p/2$ salps as leaders;

(b) the paper takes the + sign in (9) if $\xi_3 \ge 0$ for $\xi_3 \sim U(0, 1)$, which always holds, whereas the implementation compares ξ_3 with 0.5 and so takes either sign with probability 1/2;

(c) the salps are re-sorted by F_p after every iteration (Algorithm S1, line 12), so the leaders are the better half of the current population, whereas the paper only selects the best salp as the food source.

Both methods are therefore tested as published in [4].

Laplace step. Eq. (10) draws γ by inverse-CDF sampling from a Laplace distribution with location ϕ and scale χ: the sign of $\gamma - \phi$ is that of $z - \frac{1}{2}$, and $|\gamma - \phi| = -\chi \ln(1 - 2|z - \frac{1}{2}|)$ is exponential with mean χ. With $\phi = 0$, the step is centered on $\mathbf{X}^i$: a move toward $\mathbf{H}$ ($\gamma > 0$) and one away from it ($\gamma < 0$) are equally likely, and, as γ is a scalar, $\mathbf{Y}^i$ lies on the line through $\mathbf{X}^i$ and $\mathbf{H}$ (before box repair). With $\chi = 1$, $\mathrm{E}|\gamma| = 1$, i.e., the mean step equals the distance from $\mathbf{X}^i$ to $\mathbf{H}$, and $P(|\gamma| > 1) = e^{-1} \approx 0.37$, so about 18% of the candidates overshoot $\mathbf{H}$ ($\gamma > 1$); the exponential tails, heavier than Gaussian ones, give occasional long jumps that can aid exploration. The location $\phi = 0$ is that of the Laplace crossover of Deep and Thakur [14], who fix it at zero, and $\chi = 1$ is the value of the published LX-SSA [4]; we keep both untuned. An ablation designed after the main comparison separates the Laplace step from its cost and varies the distribution of γ and the scale χ (Section 12.3): on twelve cases, the step gives no advantage over the SSA midpoint, and $\chi < 1$ is worse.

S1.4 Ranking and Test Details

For the paired run-level tests, a run is scored by a prioritized composite endpoint, feasibility first and then the benchmark objective:

- a feasible run is scored by its objective,
- an infeasible run below every feasible run, and
- infeasible runs are ordered among themselves by their minimum spacing (larger is better; boundary violations are not used).

The ranking and the tests further use the following rules.

- Paired differences with $|d| \leq 10^{-9}$ are treated as zero and dropped, both in the Wilcoxon signed-rank test and in the rank-biserial correlation r_{rb}.
- Case means are rounded to 10^{-6} objective units before ranking, so that exact ties (e.g., all methods at the wake-free optimum for $N = 2$) are not broken by floating-point noise.
- Holm's correction for the run-level tests is applied within each case over the comparisons of one table; the Type I comparison table (Table 8) and the ablation table (Table 20) are different families, so the same pair of methods (e.g., PSO-VNS vs. PSO) can have slightly different W/T/L tallies in the two tables. The W/T/L counts are descriptive tallies.
- In the case-mean Wilcoxon test, a case in which only one of the two methods has at least 15 feasible runs enters as a difference in its favor that is larger than every observed difference, and a case in which neither has is dropped (Section 7).

The 68 cases are not independent (nested N within six farm and wind combinations), so all p values describe this benchmark, not a population of farms.

S1.5 Iteration Counts and Phase-1 Budgets

With $N_p = 30$ and $B = 6{,}030$, stand-alone PSO, SSA and DE run for $(B - N_p)/N_p = 200$ iterations and LX-SSA for $(B - N_p)/(2N_p) = 100$. A hybrid with split ω runs

$$T_1 = \mathrm{round}((\omega B - N_p)/c)$$

Phase-1 iterations, where c is the number of evaluations per iteration (N_p for PSO and SSA, $2N_p$ for LX-SSA) and Python's round-half-to-even rule applies; for $\omega = 0.5$ the argument is 99.5 for PSO and SSA, which is rounded to $T_1 = 100$ (3,030 evaluations), and LX-SSA gets $T_1 = 50$ (3,030 evaluations). For the split study, $\omega = 0.25$ gives 49 PSO iterations (1,500 evaluations), $\omega = 0.75$ gives 150 (4,530 evaluations) and $\omega = 0.9$ gives 180 (5,430 evaluations).

RS-VNS evaluates $\mathrm{round}(\omega B) = 3{,}015$ random layouts (including the seeded initial population of 30), about the same effort as the 3,030 Phase-1 evaluations of the swarm hybrids; its random layouts, like all random initial layouts, are uniform in the bounding square $[-r, r]^{2N}$, so all N turbines lie inside the circle only with probability $(\pi/4)^N$, and a sample that also satisfies the spacing constraint is far rarer (Table S48). Both random-sampling controls switch to VNS after these 3,015 evaluations (the swarm hybrids after 3,030). The disc-sampling control RSD-VNS evaluates the same 3,015 layouts; only its 2,985 layouts after the common initial population are drawn uniformly in the farm disc instead (Section S3).

DE/rand/1/bin (scale factor 0.5, crossover rate 0.9) clips the mutant to the box before crossover and replaces a member immediately when its trial vector is better (asynchronous one-to-one selection).

S1.6 MS-SLSQP

MS-SLSQP does not minimize F_p: it minimizes the wake loss (without penalty) under explicit inequality constraints, using SciPy's SLSQP [5] with

- the bounds $[-r, r]$ on every coordinate,
- at most 200 iterations per start,
- ftol $= 10^{-9}$ and
- forward-difference gradients with a step of 1 m.

The constraints are scaled,

$$(\ell_{ij}^2 - \ell_{\min}^2)/\ell_{\min}^2 \geq 0 \quad \text{for every pair and}$$
$$(r^2 - x_i^2 - y_i^2)/r^2 \geq 0 \quad \text{for every turbine}$$

(for Horns Rev 1, the signed distance of every turbine to every edge of the parallelogram divided by the half-width of the search box, 943 m).

The 30 initial layouts are evaluated with F_p and used as starts in increasing order of F_p; after the 30 initial layouts, every further start is a new random layout drawn from the same generator (uniform in the box, or feasible under the feasibility-preserving initialization), until the budget is used up. Every objective call, including the finite-difference calls, counts as an evaluation. MS-SLSQP returns the best evaluated layout (by wake loss) whose scaled constraints are all $\geq -10^{-9}$; the reported feasibility of this layout is then judged with the common tolerance of 10^{-6} m (Section 7). These are standard, untuned settings; since the top-hat Jensen wake makes the wake loss discontinuous in the coordinates, finite-difference gradients are unreliable, which handicaps MS-SLSQP on this benchmark.

S1.7 Feasibility-Preserving Initialization

Algorithm S2 generates each of the N_p initial layouts of the feasible-start study (Section 8.4.2); it uses no objective evaluations, so it does not count toward the budget. With spacing $s = \ell_{\min}$, it minimizes the smooth packing penalty

$$P(\mathbf{z}) = \frac{1}{\tilde{s}^4}\Big[\tfrac{1}{2}\sum_{i<j}\max\big(0, \tilde{s}^2 - \ell_{ij}^2\big)^2 + \sum_i b_i\Big], \qquad \tilde{s} = 1.0005\, s, \tag{S2}$$

with

$$b_i = \begin{cases} \max(0, x_i^2 + y_i^2 - \tilde{r}^2)^2, \quad \tilde{r} = (1 - 10^{-7})\, r, & \text{for the circular farms,} \\ \tilde{s}^2 \sum_e \max(0, \delta_{ie})^2 & \text{for the Horns Rev 1 parallelogram,} \end{cases}$$

with δ_{ie} the signed distance of turbine i outside edge e. The slightly inflated spacing and shrunken radius leave a margin, so that the result passes the feasibility test. All random draws come from the seeded global generator, so the runs of all methods with the same seed start from the same feasible initial population and stay paired; under this initialization the later random restarts of MS-SLSQP also use Algorithm S2. RS-VNS and RSD-VNS were not run with this initialization, since they would need one packing solve for each of their 3,015 random layouts.

Algorithm S2 Feasibility-preserving initialization of one layout (N turbines, spacing s, circle of radius r or convex polygon)

for try = 1 to 200 **do**
  Draw N positions uniformly in the site (circle: uniform in the disc; polygon: rejection sampling from its bounding box).
  Minimize P (S2) from these positions with L-BFGS-B (analytic gradient; at most 3,000 iterations, ftol = 10^{-15}, gtol = 10^{-12}).
  Circle: move every turbine that lies outside the circle radially back onto it.
  if every pairwise distance is $\geq s - 10^{-6}$ m and every turbine lies within 10^{-6} m of the site **then**
    return the layout.
  end if
end for
Stop with an error.

S1.8 Horns Rev 1 Model

The Horns Rev 1 block (Section 9) uses the V80 power and thrust curves and the 12-sector Weibull wind climate of PyWake's Horns Rev 1 example [6], with the sector frequencies normalized to one.

- The boundary is the parallelogram through the four corner turbines of the installed 16-turbine block, enlarged by 0.1% about the centre of the block so that the installed turbines (integer coordinates) lie inside it; in F_p (5), the boundary violation of a turbine is its distance (m) outside the parallelogram, and the spacing violation is $\ell_{\min} - \ell_{ij}$ (m) with $\ell_{\min} = 4D = 320$ m.
- The search box is the square of half-width 943 m (the largest absolute coordinate of the parallelogram) centred on the block; the same half-width is the r of the VNS radii.

- The wake model is the Jensen top-hat wake with
  - $K = 0.04$,
  - a hub-centre in-wake test (turbine i is waked by j if it lies downstream of j and within $R + Kx$ of its axis, where x is the downstream distance; there is no rotor-overlap fraction),
  - the thrust coefficient at the free-stream speed (which gives a somewhat lower wake loss than PyWake; Section S4.2) and
  - root-sum-square superposition.
- The wind speed is discretized in 1-m/s bins centred at 3, 4, ..., 25 m/s, each with the Weibull probability of its ± 0.5 m/s interval.
- The wind direction is discretized in 5° bins centred at 2.5°, 7.5°, ..., 357.5° (meteorological convention); every 30° sector (centred at 0°, 30°, ..., 330°) contains exactly six bins, each carrying the Weibull parameters of its sector and one sixth of its frequency.

Direction binning. Bins centred at 0°, 5°, ..., 355° and assigned to sectors by rounding half to even would give the sectors seven and five bins alternately, with frequencies summing to 1.003; the binning above gives every sector six bins and frequencies summing to one. All Horns Rev 1 runs used for the results (all methods, budgets and initializations), the installed and wake-free AEP values and all Horns Rev 1 results in the main text and in this document use this binning. The check of the model against PyWake is described in Section S4.2.

S1.9 Feasibility Tolerance

The penalty F_p treats any positive violation as infeasible, whereas every reported feasibility (final layouts, convergence curves, ranking rule) uses the tolerance of 10^{-6} m on both the spacing and the boundary, in all studies. Because the penalty term is $(1 + \mu_\mathrm{p} g)^2$ with $\mu_\mathrm{p} = 10^{10}$, the comparison of F_p values is a feasibility-first rule only for violations above $(\sqrt{L} - 1)/\mu_\mathrm{p}$, where L is the wake loss of the feasible layout compared with; for the benchmark cases this is at most 4.6×10^{-8} (m for the spacing, m^2 for the boundary; Lemma S6). A smaller violation adds little to F_p and can beat a feasible layout. Conversely, violations between about $5 \cdot 10^{-8}$ and 10^{-6} m are heavily penalized in F_p but counted as feasible in the reporting; the penalty's effective tolerance is thus much stricter than the reporting tolerance.

S2 Type I Farms (Kusiak–Song Benchmark): Details

This section gives the per-case results, the full case-level statistics, the coordinates of the best layouts and further figures of the Type I comparison (Section 8). These tables and figures cover the nine methods; the case-level analysis is that of Table 8 of the main text, and Section S10.2 compares GA with each other method (Table S55).

S2.1 Per-Case Results

Tables S2–S7 give the mean and standard deviation of the benchmark objective of every method in every case.

Table S2: Data Set I, 500-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 evaluations. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; "–" means no feasible run.

N	Wake-free	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	GA	VNS	MS-SLSQP
2	28091.5	**28091.5 (0.0)**	**28091.5 (0.0)**	**28091.5 (0.0)**	**28091.5 (0.0)**	**28091.5 (0.0)**	**28091.5 (0.0)**	**28091.5 (0.0)**	**28091.5 (0.0)**	28086.7 (4.5)
3	42137.2	42131.3 (8.5)	42128.7 (8.8)	42128.5 (10.7)	42114.9 (10.3)	42114.8 (14.2)	42130.7 (8.8)	42123.8 (11.5)	**42131.6 (9.7)**	42119.9 (11.0)
4	56182.9	**56114.4 (20.7)**	56110.0 (25.6)	56101.8 (25.3)	56086.8 (19.8)	56078.8 (22.6)	56092.8 (29.8)	56104.8 (19.6)	56106.1 (21.7)	56057.0 (55.8)
5	70228.7	70051.2 (28.5)	**70053.1 (29.3)**	69967.2 (159.2)	69994.0 (34.0)	69972.6 (72.0)	69960.5 (78.6)	70037.3 (23.7)	69876.9 (199.8)	69876.4 (139.3)
6	84274.4	83918.5 (66.9)	**83925.6 (55.8)**	83780.8 (213.2)	83535.6 (372.5)	83544.9 (465.3)	83326.7 (761.5)	83869.3 (145.5)	83299.3 (524.5)	83636.4 (227.9)
7	98320.2	**97588.7 (292.3)**	97569.6 (325.2)	96529.9 (794.9)	96757.0 (692.4)	96380.6 (1013.8)	95903.3 (1316.3)^{15}	97488.8 (373.8)	96967.1 (590.6)	97049.0 (477.4)
8	112365.9	110777.2 (718.5)	**110957.4 (615.4)**	109316.4 (1117.4)	109172.6 (1362.4)	108258.3 (1704.3)^{28}	107632.7 (1592.8)^{2}	110798.8 (547.2)	109493.5 (2035.2)	110233.3 (571.9)
9	126411.6	123153.2 (1132.3)	123499.5 (1006.4)	121985.3 (1201.8)	120043.1 (2755.8)^{25}	119134.2 (2666.4)^{21}	–^{0}	**123771.2 (795.4)**^{29}	122259.2 (1232.0)	123307.1 (814.5)
10	140457.4	135160.5 (1708.4)	135396.0 (1790.4)	134321.3 (1482.4)^{26}	132616.3 (2328.0)^{12}	130967.4 (3191.0)^{13}	130994.8 (0.0)^{1}	134409.4 (1352.5)^{17}	133567.5 (1820.9)	**135642.4 (920.1)**

S6 Theoretical Details

This section states and proves the theoretical facts used in the main text: the stability conditions of the implemented PSO update and the stability of its constriction setting (Section S6.1), the budget accounting of the two-phase hybrids and the strength of the random-sampling control (Section S6.2), what can and cannot be proven about PSO and DE from feasible starts (Section S6.3), the practical-equivalence statements (Section S6.4) and the property of the VNS local search (Section S6.5). All statements refer to the code in `analysis/`; all numbers of this section are recomputed and checked by `analysis/make_theory_figures.py`.

S6.1 Stability of the Implemented PSO Update

In the code, a move of particle i updates every coordinate m by

$$v \leftarrow w\,v + c_1\rho_1\,(p - x) + c_2\rho_2\,(g - x), \qquad x \leftarrow \min\big(\max(x + v, -r),\, r\big), \tag{S4}$$

where $x = X^i_m$, $v = V^i_m$, $p = P^i_m$, $g = G_m$, and ρ_1, ρ_2 are independent $U(0,1)$ numbers drawn anew for every coordinate and every move; v is not changed when x is clipped. We use two assumptions: (A1) *stagnation*: p and g are fixed; (A2) *no active bound*: the clipping never acts, so $v_t = x_t - x_{t-1}$. Under (A1)–(A2), with $\varphi_{k,t} = c_k\rho_{k,t}$ and $\varphi_t = \varphi_{1,t} + \varphi_{2,t}$, each coordinate follows, independently of the others,

$$x_{t+1} = (1 + w - \varphi_t)\,x_t - w\,x_{t-1} + \varphi_{1,t}\,p + \varphi_{2,t}\,g. \tag{S5}$$

The coefficient φ_t is random, with mean $\mu = (c_1 + c_2)/2$ and variance $\sigma^2 = (c_1^2 + c_2^2)/12$. We call the update *order-1 stable* if, for all initial values (x_0, x_1), $\mathrm{E}\,x_t$ converges to a limit that does not depend on them, and *order-2 stable* if, in addition, $\mathrm{E}\,x_t^2$ does [7]. Proposition S1 applies these known stagnation results [7] to the implemented update (summarized in Section 4.1 of the main text); it is not claimed as new theory, and its proof follows.

Proposition S1 (Stability of the implemented update) *Let $c_1, c_2 \geq 0$, $c_1 + c_2 > 0$, and assume (A1)–(A2).*

(a) $\mathrm{E}\,x_t \to x^* = (c_1p + c_2g)/(c_1 + c_2)$ for all initial values if and only if $|w| < 1$ and $0 < \mu < 2(1 + w)$, i.e.,

$$|w| < 1 \quad \textit{and} \quad c_1 + c_2 < 4(1 + w). \tag{S6}$$

(b) The update is order-2 stable if and only if $|w| < 1$ and

$$\varrho(1) := 2\mu(1 - w^2) - (1 - w)\mu^2 - (1 + w)\sigma^2 > 0. \tag{S7}$$

For $c_1 = c_2$, (S7) is Poli's condition $c_1 + c_2 < 24(1 - w^2)/(7 - 5w)$ [7]; for $c_1 \neq c_2$ it is stricter, since σ^2 grows when $c_1 + c_2$ is kept and $c_1 \neq c_2$.

(c) Under (b), $\mathrm{Var}\,x_t \to V_\infty = (1 + w)\,q/\varrho(1)$ with $q = c_1^2c_2^2(p - g)^2/[6(c_1 + c_2)^2]$; for $c_1 = c_2 = c$,

$$V_\infty = \frac{c\,(1 + w)}{4\,[12(1 - w^2) - c\,(7 - 5w)]}\,(p - g)^2, \tag{S8}$$

as in [7]. The limit is zero if and only if $c_1c_2(p - g) = 0$; for $c_1, c_2 > 0$, this reduces to $p = g$.

Proof. (a) Since $\rho_{1,t}, \rho_{2,t}$ are independent of (x_t, x_{t-1}), the deviation $e_t = \mathrm{E}\,x_t - x^*$ satisfies $e_{t+1} = m\,e_t - w\,e_{t-1}$ with $m = 1 + w - \mu$ (the constant terms cancel because $\mathrm{E}\varphi_{1,t}\,(p - x^*) + \mathrm{E}\varphi_{2,t}\,(g - x^*) = 0$). Its characteristic polynomial $\lambda^2 - m\lambda + w$ has both roots strictly inside the unit circle if and only if $|w| < 1$ and $|m| < 1 + w$ (Schur–Cohn conditions for a real monic quadratic), i.e., $0 < \mu < 2(1 + w)$; otherwise an initial value exciting a root of modulus ≥ 1 keeps e_t away from zero.

(b) Let $y_t = x_t - x^*$. Then $y_{t+1} = (1 + w - \varphi_t)y_t - wy_{t-1} + \eta_t$ with $\eta_t = \varphi_{1,t}(p - x^*) + \varphi_{2,t}(g - x^*)$, $\mathrm{E}\eta_t = 0$, and (φ_t, η_t) is independent of (y_t, y_{t-1}). Hence $z_t = (\mathrm{E}y_t^2,\ \mathrm{E}y_ty_{t-1},\ \mathrm{E}y_{t-1}^2)^{\mathsf{T}}$ satisfies $z_{t+1} = M\,z_t + b_t$ with

$$M = \begin{pmatrix} a & -2wm & w^2 \\ m & -w & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad a = \mathrm{E}(1 + w - \varphi_t)^2 = m^2 + \sigma^2,$$

and $b_t = (q - 2\,\mathrm{Cov}(\varphi_t, \eta_t)\,\mathrm{E}y_t,\, 0,\, 0)^{\mathsf{T}}$, $q = \mathrm{E}\eta_t^2$. Order-2 stability is equivalent to a spectral radius of M below one (and $|w| < 1$). Sufficiency: $\mathrm{E}y_t$ obeys the mean recursion of (a), which does not involve η_t, so it equals the mean of the process with $p = g$ (where $\eta_t = 0$ and $z_t = M^t z_0 \to 0$) and $(\mathrm{E}y_t)^2 \leq \mathrm{E}y_t^2 \to 0$ there; hence

$b_t \to (q,0,0)^\mathsf{T}$ and $z_t \to (I-M)^{-1}(q,0,0)^\mathsf{T}$, independent of the initial values. Necessity: order-2 stability includes order-1 stability, so $|w|<1$ by (a) and $\mathrm{E}y_t \to 0$; hence $\mathrm{E}y_t^2 = \mathrm{E}x_t^2 - 2x^*\mathrm{E}x_t + x^{*2}$ converges to a limit that does not depend on the initial values. The second row of M is the recursion $\mathrm{E}y_{t+1}y_t = m\,\mathrm{E}y_t^2 - w\,\mathrm{E}y_t y_{t-1}$ (the term $\mathrm{E}\eta_t\,\mathrm{E}y_t$ vanishes), which, with $|w|<1$, makes $\mathrm{E}y_t y_{t-1}$ converge whenever $\mathrm{E}y_t^2$ does (also if $wm=0$), and the third component of z_t is the first one delayed by one step; so the whole vector z_t converges to a limit that does not depend on the initial values. Now, for the initial values $(\lambda y_0, \lambda y_1)$, $\mathrm{E}y_t$ is proportional to λ, so $z_t = \lambda^2 M^t z_0 + \lambda u_t + v_t$ with u_t, v_t independent of λ; if z_t has the same limit for three values of λ, then $M^t z_0 \to 0$, and since the vectors $z_0 = (y_1^2, y_1y_0, y_0^2)$ span $\mathbb{R}^3$, $M^t \to 0$. The characteristic polynomial of M is $\varrho(\lambda) = \lambda^3 + a_2\lambda^2 + a_1\lambda + a_0$ with $a_2 = w-a$, $a_1 = 2wm^2 - aw - w^2$, $a_0 = -w^3$. By Jury's criterion, all roots lie strictly inside the unit circle if and only if (i) $\varrho(1)>0$, (ii) $\varrho(-1)<0$, (iii) $|a_0|<1$ and (iv) $|a_0^2-1| > |a_0a_2 - a_1|$. Direct expansion gives $\varrho(1) = (1+w)(1-w^2) - (1+w)\sigma^2 - (1-w)m^2$, which equals (S7), and $\varrho(-1) = -(1-w)(1-w^2+\sigma^2) - (1+w)m^2 < 0$ for $|w|<1$; (iii) is $|w|<1$. For (iv), $a_0a_2 - a_1 = w\,X$ with $X = \sigma^2(1+w^2) - (1-w^2)m^2 + w(1-w^2)$. If (i) holds, then $(1-w)m^2 < (1+w)(1-w^2)$, i.e., $m^2 < (1+w)^2$, and $\sigma^2 < 1-w^2$; therefore $-(1-w^2)(1+w+w^2) < X < (1-w^2)(1+w+w^2)$, and $|wX| < |w|(1+w+w^2)(1-w^2) \le (1+w^2+w^4)(1-w^2) = 1-w^6$, because $1+w^2+w^4 - |w|(1+w+w^2)$ equals $(1-w)(1-w^3)$ for $w \ge 0$ and $1-u+2u^2-u^3+u^4 > 0$ for $w=-u<0$. So (iv) follows from (i), and order-2 stability holds if and only if $|w|<1$ and $\varrho(1)>0$. For $c_1 = c_2 = c$: $\mu = c$, $\sigma^2 = c^2/6$ and $\varrho(1) = c\,[12(1-w^2) - c(7-5w)]/6$, which is positive if and only if $2c < 24(1-w^2)/(7-5w)$. For fixed c_1+c_2, σ^2 is smallest at $c_1=c_2$, which gives the last statement of (b). (The general condition restates, in our notation, the second-moment analysis of [7] for $c_1 \ne c_2$; we do not claim it as new.)

(c) In the limit $\mathrm{E}y_t \to 0$, so $b_t \to (q,0,0)^\mathsf{T}$ and $z_t \to z_\infty$ with $z_\infty = Mz_\infty + (q,0,0)^\mathsf{T}$. The second row gives $\mathrm{E}y_t y_{t-1} \to mV_\infty/(1+w)$, and the first row then gives $V_\infty\,\varrho(1)/(1+w) = q$. With $p - x^* = c_2(p-g)/(c_1+c_2)$ and $g - x^* = -c_1(p-g)/(c_1+c_2)$, $q = [c_2^2\,\mathrm{Var}\,\varphi_1 + c_1^2\,\mathrm{Var}\,\varphi_2](p-g)^2/(c_1+c_2)^2 = c_1^2c_2^2(p-g)^2/[6(c_1+c_2)^2]$, which is $c^2(p-g)^2/24$ for $c_1=c_2=c$ and gives (S8). □

Which φ enters the order-1 condition. For constant coefficients, the classical condition is $0<\varphi<2(1+w)$ with $\varphi = c_1+c_2$. In the implemented update the coefficients are $c_k\rho_k$ with $\rho_k \sim U(0,1)$, and the order-1 condition involves their mean, $\mathrm{E}\varphi_t = (c_1+c_2)/2$; hence $c_1+c_2 < 4(1+w)$. The analytic order-2 region of Proposition S1(b) agrees with the numerically computed spectral radius of M at 200,000 random points $(w,c_1,c_2) \in (-1,1)\times[0,6]^2$ and on a 601×601 grid for $c_1=c_2$ (points within 10^{-3} of the boundary excluded; no disagreement).

Table S47: Stability of the constriction setting of PSO under stagnation and without active bounds (Proposition S1). ρ_1: spectral radius of the mean recursion; ρ_2: spectral radius of the second-moment matrix M (order-2 stable iff $\rho_2<1$); $V_\infty/(p-g)^2$: limit variance factor (S8).

Setting	w	c_1+c_2	$4(1+w)$	$\frac{24(1-w^2)}{7-5w}$	margin	ρ_1	ρ_2	$V_\infty/(p-g)^2$
Constriction	0.7298	2.99	6.92	3.35	+0.36	0.85	0.94	1.09

Margin: order-2 bound minus c_1+c_2. Constriction: $c_1+c_2 = 2.99236$ is 10.6% below the bound 3.3475.

Proposition S2 (The constriction setting) *Under (A1)–(A2), the constriction setting is order-1 stable and order-2 stable ($c_1+c_2 = 2.99 < 3.35$, $\rho_2 = 0.94$); each coordinate of a particle with $p \ne g$ has, in the limit, mean $(p+g)/2$ and standard deviation $1.04\,|p-g|$ ($V_\infty = 1.09\,(p-g)^2$).*

Proof. The numbers follow from Proposition S1 (Table S47); the limit mean is $x^* = (p+g)/2$ because $c_1=c_2$. □

What the conditions do not cover. (i) *Stagnation.* In a run, $\mathbf{P}^i$ and $\mathbf{G}$ change. For $c_1=c_2$, Cleghorn and Engelbrecht [17] obtain the same region under the weaker non-stagnate distribution assumption; we do not claim that their region extends to $c_1 \ne c_2$, where the condition under stagnation is stricter (Proposition S1(b)). No stability result says how good $\mathbf{G}$ is: order-2 stability means that each particle samples a distribution of bounded variance around $(c_1\mathbf{P}^i + c_2\mathbf{G})/(c_1+c_2)$, whose variance vanishes only if $\mathbf{P}^i = \mathbf{G}$ (Proposition S1(c)); it implies neither convergence to a point nor local optimality. (ii) *Bounds.* The code clips the position to $[-r,r]$ and keeps the velocity, so after a clip $v_t \ne x_t - x_{t-1}$, and (S5) and both results no longer describe the trajectory. What the code implies at a bound is the following.

Lemma S3 (Boundary handling of the PSO code) *(a) If a move leaves a coordinate of turbine k at $\pm r$, then $x_k^2 + y_k^2 \ge r^2$, with equality only if its other coordinate is exactly zero; so the evaluated layout violates the boundary constraint of the circular farm unless that coordinate is zero. (b) If a coordinate is at r after a move and its kept velocity is $v \ge 0$, the next velocity is $v' = wv + \varphi_1(p-r) + \varphi_2(g-r) \le wv$, and the coordinate is clipped to r again as long as $v' \ge 0$; the same holds at $-r$ with the signs reversed.*

Proof. (a) $x_k^2 + y_k^2 = r^2 + y_k^2$ if $x_k = \pm r$ (and likewise for y_k). (b) p and g are evaluated, hence clipped, positions, so $p - r \le 0$ and $g - r \le 0$; the position update gives $\min(r + v', r) = r$ for $v' \ge 0$. □

Thus, in the implementation, a coordinate that reaches the bound stays there while its velocity points outward, and by (a) every such evaluation is infeasible (almost surely). Lemma S3 does not say how often clipping occurs; that is an empirical question, answered by the diagnostics of Section S7.1.

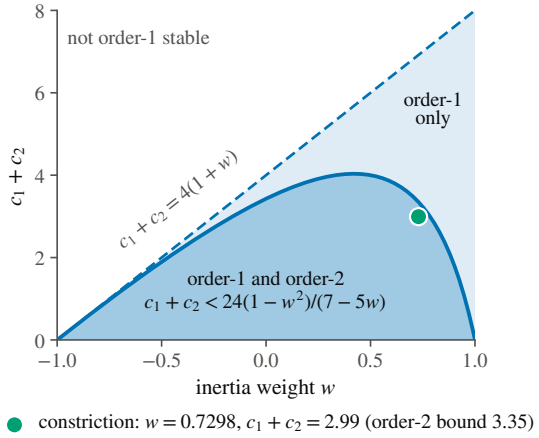


Figure S11: Stability regions of the implemented PSO update under stagnation and without active bounds ($c_1 = c_2$; Proposition S1). Light: order-1 stable only, $c_1 + c_2 < 4(1 + w)$; dark: order-1 and order-2 stable, $c_1 + c_2 < 24(1 - w^2)/(7 - 5w)$. The constriction setting (8) (dot) is order-2 stable.

S6.2 Budget Accounting and the Random-Sampling Control

An *evaluation* is one call of the objective, counted by a wrapper that stops the run at the budget B (for MS-SLSQP also the finite-difference calls). A population method with N_p members uses N_p evaluations for the initial population and c per iteration, $c = N_p$ for PSO, SSA and DE and $c = 2N_p$ for LX-SSA, so T iterations cost $N_p + cT$ evaluations; stand-alone, $T = \lfloor (B - N_p)/c \rfloor$ (Section S1.5). A two-phase hybrid with split ω uses $B_1 = N_p + cT_1$ evaluations in Phase 1, $T_1 = \mathrm{round}((\omega B - N_p)/c)$ with Python's rounding half to even (for PSO and SSA at $\omega = 0.5$, 99.5 is rounded to $T_1 = 100$), and $B_2 = B - B_1$ in Phase 2; RS-VNS and RSD-VNS use $B_1 = \mathrm{round}(\omega B)$, including the seeded initial population. At $B = 6{,}030$ and $\omega = 0.5$, $B_1 = 3{,}030$ for the swarms and 3,015 for the two random-sampling controls, a difference of 15 evaluations (0.25% of B).

Proposition S4 (Prefix and monotonicity) *For every seed: (a) Phase 1 of PSO-VNS evaluates exactly the first B_1 layouts of the stand-alone PSO run; (b) the final F_p of PSO-VNS is at most the best F_p of these B_1 evaluations; (c) (a) does not hold for SSA-VNS and LX-SSA-VNS.*

Proof. (a) Both runs seed the global generator once and then draw, in the same order, the initial population and the vectors $\boldsymbol{\rho}_1, \boldsymbol{\rho}_2$ of every move; the PSO update does not depend on the number of iterations, so the first T_1 iterations coincide. (b) Phase 2 starts with $\mathbf{G}$ as its best evaluated point, replaces it only by strictly better evaluated points and returns the best one. (c) The SSA coefficient $\xi_1 = 2\exp[-(4t/T)^2]$ depends on t/T, and the hybrid uses $T = T_1$ instead of the stand-alone iteration count. □

Why equal budgets matter. Let $L_A(b)$ be the best F_p found by method A within b evaluations. For a fixed seed, $L_A(b)$ is nonincreasing in b, so any comparison at unequal budgets mixes the effect of the method with that of the budget: counting only Phase-2 evaluations, or counting an LX-SSA iteration as N_p instead of $2N_p$ evaluations, would give one method more evaluations and an advantage by monotonicity alone. At the same B, Proposition S4 makes PSO-VNS versus PSO a clean contrast: both continue from the identical state after B_1 evaluations, PSO with its whole swarm and VNS from $\mathbf{G}$ alone, for the same B_2 evaluations. It guarantees $L_{\text{PSO-VNS}}(B) \le L_{\text{PSO}}(B_1)$ but not $L_{\text{PSO-VNS}}(B) \le L_{\text{PSO}}(B)$; which of the two continuations is better is an empirical question. Likewise, the contrast between a hybrid and RS-VNS isolates Phase 1 (a search method versus uniform sampling with about the same number of evaluations, followed by the same Phase 2). How weak this control is follows from Proposition S5, which bounds the probability P_N that a layout drawn uniformly in the bounding square is feasible. Its bound (b) uses the area $A(\rho)$ of the part of the farm disc D (radius r) covered by a disc of radius ρ centred on the boundary of D; for $0 < \rho \le 2r$,

$$A(\rho) = r^2 \arccos\Big(1 - \frac{\rho^2}{2r^2}\Big) + \rho^2 \arccos\frac{\rho}{2r} - \frac{\rho}{2}\sqrt{4r^2 - \rho^2}. \tag{S9}$$

Proposition S5 (Random sampling of layouts) *Let the $2N$ coordinates be independent and uniform in $[-r, r]$, and let P_N be the probability that the layout is feasible (all turbines in the disc $\mathcal{D}$ of radius r and all pairwise distances $\ge s$). Then (a) all N turbines lie in $\mathcal{D}$ with probability $(\pi/4)^N$; (b) $P_N \le$*

Table S48: Probability P_N that a layout drawn uniformly in the bounding square $[-r, r]^{2N}$ is feasible (Proposition S5): all turbines in the farm, $(\pi/4)^N$; rigorous upper bound (b); Monte Carlo estimate from 10^7 layouts per row (fixed seed; turbines drawn uniformly in the disc, multiplied by $(\pi/4)^N$; 95% Clopper–Pearson interval); expected number of feasible layouts among the 3,015 uniform layouts of the Phase 1 of RS-VNS at $B = 6{,}030$ (including the initial population). Rows $N = 10$, 12, 15: largest benchmark cases ($s = 8R = 308$ m); $N = 16$, 36: IEA37 Case Study 1 ($s = 2D = 260$ m); $N = 20$: hypothetical.

N	r (m)	s (m)	$(\pi/4)^N$	bound on P_N	P_N (Monte Carlo) [95% CI]	expected feasible of 3,015
2	500	308	0.62	0.52	0.44 [0.44, 0.44]	1,337
10	500	308	0.089	$6.5 \cdot 10^{-3}$	0 [0, $3.3 \cdot 10^{-8}$]	$< 9.9 \cdot 10^{-5}$
12	750	308	0.055	0.012	$9.4 \cdot 10^{-8}$ [$5.5 \cdot 10^{-8}$, $1.5 \cdot 10^{-7}$]	$2.8 \cdot 10^{-4}$
15	1,000	308	0.027	$7.0 \cdot 10^{-3}$	$3.5 \cdot 10^{-7}$ [$2.9 \cdot 10^{-7}$, $4.1 \cdot 10^{-7}$]	$1.0 \cdot 10^{-3}$
16	1,300	260	0.021	0.011	$1.5 \cdot 10^{-4}$ [$1.5 \cdot 10^{-4}$, $1.5 \cdot 10^{-4}$]	0.45
20	1,000	308	$8.0 \cdot 10^{-3}$	$7.0 \cdot 10^{-4}$	0 [0, $2.9 \cdot 10^{-9}$]	$< 8.9 \cdot 10^{-6}$
36	2,000	260	$1.7 \cdot 10^{-4}$	$4.3 \cdot 10^{-5}$	$1.8 \cdot 10^{-9}$ [$1.5 \cdot 10^{-9}$, $2.2 \cdot 10^{-9}$]	$5.4 \cdot 10^{-6}$

For $N = 2$ the exact value, from the distance distribution of two uniform points in a disc, is $P_2 = 0.443$. "<": upper end of the 95% interval (no feasible layout among the 10^7).

$(\pi/4)^N \prod_{k=1}^{N-1} \max\{0,\ 1 - \max\{A(s),\, k\,A(s/2)\}/(\pi r^2)\}$*, where* $A(\rho)$ *is the area of the part of* $\mathcal{D}$ *covered by a disc of radius* ρ *centered on the boundary of* $\mathcal{D}$ *(closed form:* (S9)*); (c) among* M *such layouts, the expected number of feasible ones is* MP_N*.*

Proof. (a) The turbines are independent and uniform in the square, each in $\mathcal{D}$ with probability $\pi r^2/(2r)^2$. (b) Given (a), the positions $\mathbf{p}_1, \ldots, \mathbf{p}_N$ are independent and uniform in $\mathcal{D}$. If $\mathbf{p}_1, \ldots, \mathbf{p}_k$ are pairwise at least s apart, $\mathbf{p}_{k+1}$ must avoid $U_k = \mathcal{D} \cap \bigcup_{j \le k} B(\mathbf{p}_j, s)$. The area of $B(\mathbf{c}, \rho) \cap \mathcal{D}$, $\mathbf{c} \in \mathcal{D}$, is nonincreasing in $\|\mathbf{c}\|$ (its derivative is minus the length of the common chord), hence $\ge A(\rho)$; so $|U_k| \ge A(s)$, and, as the discs $B(\mathbf{p}_j, s/2) \subset B(\mathbf{p}_j, s)$ are disjoint, $|U_k| \ge kA(s/2)$. The chain rule over k gives (b). (c) Linearity of expectation. □

The bound (b) is rigorous but loose; the Monte Carlo estimates show that for the largest benchmark cases a uniform layout in the square is feasible with probability below 10^{-6}, so the expected number of feasible layouts in the Phase 1 of RS-VNS is about 10^{-3} or less (Table S48). Since any violation above 4.6×10^{-8} makes F_p exceed every wake loss (Section S1.9 and Lemma S6), and $F_p = L + \mathrm{Pen}$ with $0 \le L \le NE_{\mathrm{ideal}} + \varepsilon$ (ε: the small contribution of the negative part of the power curve just above cut-in), the Phase-1 result of RS-VNS in these cases is almost surely the sampled layout with the smallest F_p, which is decided by the penalty: $F_p(\mathbf{X}) < F_p(\mathbf{Y})$ whenever $\mathrm{Pen}(\mathbf{Y}) - \mathrm{Pen}(\mathbf{X}) > NE_{\mathrm{ideal}}$. RS-VNS thus hands VNS the least-violating of its samples, not a layout selected for low wake loss, and "better than RS-VNS" is a minimum bar. The factor $(\pi/4)^N$ explains only a small part of this weakness: the Monte Carlo column of Table S48 lies orders of magnitude below $(\pi/4)^N$, so the spacing constraint dominates, and in the runs, disc sampling gives a feasible sample to only a minority of the RS-VNS runs without one (Section S3).

S6.3 Feasible Starts: What Is Proven and What Is Observed

With the feasibility-preserving initialization (Algorithm S2), all N_p initial layouts are feasible, and $\mathbf{G}$ is the best of them. From feasible starts, in all runs of the study (six largest cases and Horns Rev 1, 30 seeds each), $\mathbf{G}$ never moved (Section 8.4.2). The following facts follow from the code; the stall itself does not, and is an empirical finding. Lemma S6 and parts (a) and (b) of Proposition S7 are summarized in Section 8.4.2 of the main text; they are given here in full, together with part (c) and the move structure of DE (Proposition S8).

Lemma S6 (Improvement needs a feasible trial) *If* $\mathbf{G}$ *is feasible, a trial* $\mathbf{X}$ *replaces it only if* $L(\mathbf{X}) + \mathrm{Pen}(\mathbf{X}) < L(\mathbf{G})$*; since every violated constraint adds more than* 1 *and a violation* g *adds* $(1 + \mu g)^2$*, every violation of* $\mathbf{X}$ *must be below* $(\sqrt{L(\mathbf{G})} - 1)/\mu$*, i.e., below* 4.6×10^{-8} *(m or* m^2*) in the benchmark cases, and* $\mathbf{X}$ *must have a lower wake loss than* $\mathbf{G}$*. So* $\mathbf{X}$ *need not be exactly feasible, but its violations must be far below the reporting tolerance of* 10^{-6} *m.*

Proof. Definition of F_p (5), $\mathrm{Pen} \ge 0$ and $L \ge 0$; for the benchmark cases $L(\mathbf{G}) \le NE_{\mathrm{ideal}} + \varepsilon$ with $N \le 15$ and $E_{\mathrm{ideal}} \le 14{,}045.74$ (Data Set I), so $L(\mathbf{G}) < 2.11 \cdot 10^5$, $\sqrt{L(\mathbf{G})} < 459.1$, and with $\mu = 10^{10}$ the bound is below 4.6×10^{-8}. □

Proposition S7 (PSO from feasible starts) *Let all initial layouts be feasible and let* i^* *be the particle with the best initial layout, so that* $\mathbf{X}^{i^*} = \mathbf{P}^{i^*} = \mathbf{G}$ *and* $\mathbf{V}^{i^*} = \mathbf{0}$*.*

(a) *As long as* $\mathbf{G}$ *does not change, particle* i^* *keeps* $\mathbf{V}^{i^*} = \mathbf{0}$ *and* $\mathbf{X}^{i^*} = \mathbf{G}$*: it re-evaluates* $\mathbf{G}$ *once per iteration (200 of the 6,030 evaluations if* $\mathbf{G}$ *never changes) and evaluates no other point.*

(b) *The first move of every other particle is* $\mathbf{X}^i \leftarrow \mathbf{X}^i + c_2\boldsymbol{\rho}_2 \circ (\mathbf{G} - \mathbf{X}^i)$*, clipped to the box: every coordinate moves a random fraction in* $[0, c_2] = [0, 1.50]$ *of the way to the same coordinate of* $\mathbf{G}$*, i.e., turbine* k *of layout* i *is moved toward turbine* k *of* $\mathbf{G}$*, although the numbering of the turbines in two independently packed layouts is arbitrary.*

(c) *If no evaluation improves any* $\mathbf{P}^i$*, the swarm is in exact stagnation, and, as long as no coordinate is clipped, Propositions S1(c) and S2 apply: every coordinate of particle* $i \neq i^*$ *is sampled, in the limit, with mean* $(P_m^i + G_m)/2$ *and standard deviation* $1.04\,|P_m^i - G_m|$*, so the samples do not concentrate at* $\mathbf{G}$*.*

Proof. (a) By induction: with $\mathbf{V} = \mathbf{0}$ and $\mathbf{P}^{i^*} = \mathbf{X}^{i^*} = \mathbf{G}$, (6) gives $\mathbf{V} = \mathbf{0}$, the position stays $\mathbf{G}$ (inside the box), and its value equals $F_p(\mathbf{G})$, which is not a strict improvement, so neither $\mathbf{P}^{i^*}$ nor $\mathbf{G}$ changes. (b) $\mathbf{V} = \mathbf{0}$ and $\mathbf{P}^i = \mathbf{X}^i$ at the start. (c) Stagnation is (A1), and no clipping is (A2). □

Part (c) is illustrative only: with the typical distance of the feasible starts to $\mathbf{G}$ (median matched RMS turbine distance 291 m, Table S50), a standard deviation of $1.04\,|P_m^i - G_m|$ would place many samples outside the box, so (A2) fails and this limit is not reached in the runs.

Proposition S8 (Move structure of DE/rand/1/bin) *In the DE of the code, the trial of member* i *takes coordinate* j *from the mutant* $\mathbf{x}_{r_1} + F(\mathbf{x}_{r_2} - \mathbf{x}_{r_3})$ *($r_1, r_2, r_3 \neq i$ distinct; clipped to the box) if* $j = j_{\text{rand}}$ *or* $U_j < \text{CR}$*, and from* $\mathbf{x}_i$ *otherwise. The number* K *of mutant coordinates is* $1 + \text{Bin}(n-1, \text{CR})$*, with mean* $1 + \text{CR}(n-1)$*, e.g., 27.1 of the* $n = 30$ *coordinates for* $N = 15$ *(*$\text{CR} = 0.9$*). A turbine whose two coordinates both come from the mutant is placed at* $\mathbf{x}_{r_1,k} + F(\mathbf{x}_{r_2,k} - \mathbf{x}_{r_3,k})$*, a combination of the* k*th turbines of three other layouts. If no trial is accepted, the population keeps its initial layouts and every mutant comes from the fixed set of* $N_p(N_p-1)(N_p-2)$ *vectors* $\mathbf{x}_a + F(\mathbf{x}_b - \mathbf{x}_c)$*.*

Proof. The binomial crossover loop and the one-to-one replacement of the code. □

For comparison, the SSA leaders sample $H_m + U(-\xi_1 r, \xi_1 r)$ independently for every coordinate (the code's step $\xi_1((ub - lb)\xi_2 + lb) = \xi_1 r(2\xi_2 - 1)$ with a random sign), with $\xi_1(t) = 2\exp[-(4t/T)^2] \leq 0.037$ for $t \geq T/2$ (0.0366 at $t = T/2$); the VNS local search changes one coordinate of the incumbent by $\pm h$, and its shaking perturbs all coordinates of the incumbent. These moves perturb one layout and keep the numbering of its turbines, whereas the moves of Propositions S7(b) and S8 combine coordinates of different, independently numbered layouts. That these combined moves almost never yield a feasible layout with lower wake loss, which by Lemma S6 is the only way to move $\mathbf{G}$, is the plausible mechanism of the stall, but it is not proven here; the diagnostics measure it (share of feasible trials, violation type, personal-best updates, accepted DE trials; Section S7.3 and Table S50).

S6.4 Practical Equivalence

Practical equivalence is defined in Definition 1 of the main text. In the notation of this section, with $d_c = \bar{L}_{A,c} - \bar{L}_{B,c}$ (pp of wake loss) the difference of the case means of methods A and B in case c, over the n cases in which both methods have at least 15 of 30 feasible runs, and Δ their mean, A and B are practically equivalent at margin δ (the margin m of the main text) if the two one-sided tests (TOST) reject both $H_0^-: \Delta \leq -\delta$ and $H_0^+: \Delta \geq \delta$ at $\alpha = 0.05$, i.e., reject $H_0: |\Delta| \geq \delta$ in favor of $H_1: |\Delta| < \delta$; the margin is $\delta = 0.05$ pp, a convention fixed after the primary analysis. Because $H_0 = H_0^- \cup H_0^+$ is rejected only if both parts are, TOST has size at most α for a single method pair if each constituent one-sided test is calibrated at level α (intersection–union principle). This does not establish calibration of bootstrap tail shares, or remove multiplicity across method pairs. A null simulation at the margin therefore checks the bootstrap rule (Section S11.3 and Table S66): its empirical size is 4.4–7.4% for PSO-VNS vs. PSO and 4.7–6.0% at the seed level for SSA-VNS vs. RSD-VNS, but 15.0% at the case level for the non-superiority side of the latter (one outlying case); an empirical shifted-null check on these data gives $p = 0.015$ and $p = 0.007$ for the two case-level claims, a local diagnostic rather than a general guarantee of level-α control (Section S11.3).

Lemma S9 (Confidence-interval form) *With the* t *statistic, TOST at level* α *rejects* H_0 *if and only if the* $(1-2\alpha)$ *confidence interval* $\bar{d} \pm t_{1-\alpha,n-1}\,\text{se}$ *lies inside* $(-\delta, \delta)$*; for* $\alpha = 0.05$ *this is the 90% interval. Equivalence then holds for every margin larger than* $\delta_{\min} = \max(|\textit{lower end}|, |\textit{upper end}|)$ *and for none smaller.*

Proof. H_0^- is rejected iff $(\bar{d} + \delta)/\text{se} > t_{1-\alpha,n-1}$, i.e., iff the lower end exceeds $-\delta$; H_0^+ is rejected iff the upper end is below δ. Both ends do not depend on δ, which gives the statement on $\delta_{\min}$. □

The pipeline uses the percentile-bootstrap analogue (`analysis/mpce_results.py`, function `tost`): equivalence holds if the 90% percentile interval of the 10,000 bootstrap means (cases resampled with replacement) lies inside

$(-\delta, \delta)$, the reported p value is the larger of the two one-sided bootstrap tail shares (with the add-one correction; an inversion of the percentile interval rather than a p value calibrated at the null boundary), and $\delta_{\min}$ is reported for every pair, so readers can apply their own margin. With 10,000 resamples, the interval rule and the rule $p \leq 0.05$ can differ only in the boundary case in which exactly 500 bootstrap means lie at or beyond one of the margins (interpolation of the 5% quantile); for the 18 pairs they agree at every level. As the cases are not independent (Section S1.4), the interval describes this benchmark, not a population of farms.

Levels of inference. The same interval rule is applied at three levels, which answer different questions (Definition 1 of the main text and Table S15): at the *seed level*, the 68 cases are fixed and only the run-to-run variability is resampled (the runs within every case), which describes the fixed benchmark; at the *case level*, the cases are resampled as if exchangeable (Table S13); at the *cluster level*, a cluster-robust interval over the six (data set, radius) clusters describes dependence within the six designed benchmark groups. The reported minimal margins grow here, so an equivalence statement must name its level; for PSO-VNS vs. PSO equivalence at ± 0.05 pp holds at the seed and case levels but is not shown at the cluster level (Section S3).

One-sided statements. "A gives no gain larger than δ over B" (non-superiority of A) needs only one of the two tests of TOST: it holds at level α if the lower end of the $(1 - 2\alpha)$ interval of $\Delta = \bar{L}_A - \bar{L}_B$ lies above $-\delta$, whatever its upper end. This is the form in which we state that the SSA phase adds nothing of practical size beyond sampling in the disc (SSA-VNS vs. RSD-VNS), where two-sided equivalence is not shown.

Non-significance is not equivalence. A test of $\Delta = 0$ that does not reject only says that zero lies in its confidence interval; the interval can still extend far beyond $\pm\delta$, for example when the differences vary much between cases or there are few of them. Conversely, a difference can be significant and practically negligible. The two questions give four outcomes:

	90% CI inside $(-\delta, \delta)$	90% CI not inside $(-\delta, \delta)$
95% CI contains 0	equivalent, no significant difference	inconclusive
95% CI excludes 0	significant but practically equivalent	significant, not equivalent

Only the left column supports statements such as "as good as"; a non-significant result in the right column ("inconclusive") does not, and "no gain larger than δ" needs the one-sided form below. (The case-mean Wilcoxon test concerns a median-type location of the d_c, the TOST interval their mean; the table applies to each test with its own interval.)

Bayesian signed-rank test with a region of practical equivalence (ROPE). The Bayesian test of the pipeline [10] (function `bayes_signrank`) places a Dirichlet-process prior (strength 0.5, pseudo-observation at 0) on the distribution of the d_c and, for every posterior sample, computes the probabilities θ_A, θ_{rope}, θ_B that the average of two case differences, $(d_c + d_{c'})/2$, lies below $-\delta$, inside $[-\delta, \delta]$ or above δ. The reported $P(\text{rope})$ is the posterior probability that θ_{rope} is the largest of the three (likewise $P(A \text{ better})$, $P(B \text{ better})$). It is thus (i) a posterior probability given the prior and the exchangeability of the cases, not an error rate; (ii) a statement about the typical pairwise-averaged case difference, not about the mean Δ of TOST, so the two can disagree, e.g., when most cases differ little but a few differ much; and (iii) a Monte Carlo share, so a reported value of 1.00 means that θ_{rope} was largest in (almost) every posterior sample, not certainty. We therefore report it as the posterior probability that practical equivalence is the most probable of the three outcomes, together with the posterior means of θ_A, θ_{rope} and θ_B (Table S25); like the case-level TOST, it treats the cases as exchangeable.

S6.5 The Local Search of VNS

The local search of VNS (Section 4.3) is a compass search: from $\mathbf{y}$ with step h it evaluates the $2n$ points $\text{clip}(\mathbf{y} \pm h\mathbf{e}_m)$, moves to the best one if it strictly improves F_p, and halves h otherwise, starting at $h_0 = 0.05r$ and stopping when $h < h_{\min} = 10^{-3}r$. Shaking draws a point uniformly in the ℓ_∞ shell $\delta_{k-1} < \|\mathbf{X}' - \mathbf{X}\|_\infty \leq \delta_k$, $\delta_k = 0.1kr$ (by rejection), and then clips it to the box; the clipped point is therefore neither uniform on the shell nor necessarily in it.

Proposition S10 (Coordinate-wise local minimality) *Unless the budget ends first, the local search stops after finitely many evaluations at a point* $\mathbf{y}$ *with* $F_p(\mathbf{y}) \leq F_p(\mathbf{y}_0)$ *and*

$$F_p\big(\text{clip}(\mathbf{y} \pm \bar{h}\,\mathbf{e}_m)\big) \geq F_p(\mathbf{y}) \quad \textit{for all } m = 1, \ldots, n, \qquad \bar{h} = h_0/2^5 = 0.0015625\,r. \tag{S10}$$

In PSO-VNS, if B_2 suffices for the first descent, the Phase-2 incumbent therefore satisfies (S10) *and $F_p \leq F_p(\mathbf{G})$, and so does every later incumbent, since each results from a completed descent.*

Proof. For a fixed h, every accepted move strictly decreases F_p, so no point is visited twice, and every coordinate of a visited point lies in the finite set $\{a + kh \colon k \in \mathbb{Z}\} \cap [-r, r]$ with a the value of that coordinate at the start of this h, $-r$ or r; hence each h ends after finitely many moves, and there are six values of h. The loop ends

only by halving the step to below $h_{\min}$, which happens only after a complete sweep with no strict improvement at the current step $\bar{h}$; $\bar{h} = h_0/2^5$ because $0.05/2^5 = 0.0015625 \geq 10^{-3} > 0.05/2^6$. □

Condition (S10) is a coordinate-wise optimality at one step length: $\bar{h}$ is 0.78 m for $r = 500$ m and 1.56 m for $r = 1000$ m. It does not imply optimality for smaller or larger steps, for moves of several coordinates, or a local minimum in the usual sense; F_p is discontinuous (top-hat wake), so no stationarity statement is available either. It does not require a converged PSO start; the start only sets $F_p(\mathbf{y}_0)$. If the budget ends during a descent, the returned layout (the best one evaluated) has F_p no larger than the incumbent but no such certificate. At $B = 6{,}030$ this is the rule, not the exception, for the larger cases: for $N \geq 12$ the first descent of PSO-VNS is unfinished in 39 of the instrumented runs (Table S49), so Proposition S10 gives no certificate for the largest layouts; with best-improvement polling, one move costs $2n$ evaluations.

S7 Algorithm Diagnostics

This section looks inside the runs: how often PSO clips coordinates to the box (Section S7.1), what happens in the VNS phase of the hybrids (Section S7.2), why PSO and DE do not move from feasible starts (Section S7.3) and the Phase 1 of RS-VNS (Section S7.4). The data come from instrumented reruns of subsets of the runs of the paper (`analysis/mpce_diagnostics.py`). The instrumented optimizers are verbatim copies of the code of the paper with logging lines that only read the state and draw no random numbers. All 522 instrumented runs reproduce the stored runs of the paper (same wake loss, coordinates, convergence curve and number of evaluations; 522 of 522), and seed 1 of every case and method, rerun with the original, uninstrumented code in the same process, gives bit-identical results (62 of 62 runs).

S7.1 Clipping in PSO

How often the clipping of Lemma S3 occurs was measured in instrumented reruns of PSO on 6 cases × 10 seeds (6,030 evaluations, 200 iterations; a clipped coordinate keeps its velocity): 1.6% of the coordinates leave the bounding square and are clipped per iteration (iterations 1–200), 48.4% of the particle evaluations in iterations 101–200 are feasible, and the final global best is feasible in 60 of the 60 runs. Of the 2040 stored PSO runs on the 68 benchmark cases, 2 end with a coordinate on the box bound, and 0 have an infeasible final layout.

S7.2 Inside the VNS Phase

Table S49 follows Phase 2 of PSO-VNS, SSA-VNS and RS-VNS on the 12 cases of the split study × 10 seeds. At 6,030 evaluations, Phase 2 is essentially one compass-search descent: for PSO-VNS, the first descent from the switch point takes a median of 2,860 evaluations (pooled share 78.5% of the Phase-2 evaluations; median share per run 95.3%) and yields 97.5% of the Phase-2 improvement, and 114 of 120 runs have no accepted shaking cycle. A shake costs a single evaluation; the relevant cost is that of the cycles (shake plus the local search that follows it), which take 21.5% of the Phase-2 evaluations (pooled; median run 4.7%) and give 0.5% of the Phase-2 gain (2.5% including the cycle cut off by the budget). For $N \geq 12$ the median run has no shaking step at all, because the first descent almost never finishes within the budget, and all accepted cycles use the smallest neighbourhood $k = 1$. VNS never makes a feasible incumbent infeasible, and its start is always the best feasible Phase-1 layout when Phase 1 evaluated one. The component analysis (Section 12) therefore measures the value of a local-search phase after Phase 1, a truncated best-improvement compass search at this budget; the neighbourhood structure of VNS can matter only for small N, and the local-minimality certificate of Proposition S10 does not apply to the runs in which the first descent is unfinished.

S7.3 Feasible Starts

Table S50 logs every candidate of PSO and DE from feasible starts (7 cases × 3 seeds). Neither method improves on the best initial layout, in these runs or in any of the 210 stored runs of each. Of the 126,000 PSO candidates, 3.33% are feasible, and every one of them is the particle holding $\mathbf{G}$ (with zero velocity) re-evaluating $\mathbf{G}$, as in Proposition S7(a); no personal best is ever updated, and none of the 126,000 DE trials is feasible or accepted. Label symmetry does not explain the stall: after the turbines of each start are relabelled to their nearest turbines of $\mathbf{G}$, the first PSO move is still feasible in only 0.1% of the draws, and only a common weight for all coordinates (a convex combination of two layouts) gives 16.6%. The feasible starts are distinct dense layouts (median RMS turbine distance to $\mathbf{G}$ after relabeling 291 m), and the independent per-coordinate weights, with overshoot beyond $\mathbf{G}$, place turbines too close to each other or outside the site.